COUNTRY STRATEGY BRIEF

France and the Allied Industrial Base

What to secure, build, source from partners, and deploy

Generated from Allied Industrial Base Explorer data; evidence, sources and caveats in the appendix. Data as of 30 October 2026.

1. Bottom line — France's strategic role

France's distinctive role in the Allied Industrial Base is as its nuclear specialist and a broad, mid-sized equipment maker. Allies rely on it for reactor parts, 20% of world exports or production, and it holds core seats in Nuclear Fuel & Reactor Systems and Electric Vehicles. Elsewhere its roles are substitutable, and the AIB assigns it no primary strategic role at its current thresholds. Its exposure is imported fuel and the control of imported equipment: oil and gas, grid and battery software, rare-earth magnets and transformer steel sit behind the High fence. Its industrial effort should keep enrichment, wind parts and battery components competitive and catch up in EVs, drivetrains and battery cells, where allies also need new capacity. Most of its manufacturing positions are a few percent of world supply, so catching up should go with allied makers. The nuclear manufacturing calls need care, because their trade baskets mix reactor parts with boiler and steam-turbine parts. At home only EV adoption needs a faster build-out, and France's own need gives its development cooperation a narrow base, battery cells.[1,2]

SECURE

Oil and gas supply, which takes one to three years to rebuild; potentially coercive suppliers provide 51% of oil imports. The control software of battery, grid and inverter equipment. Rare-earth magnets for EV motors, where China leads world rare-earth refining, and grain-oriented electrical steel for transformers. The answer is trusted contracts, stocks, qualified suppliers and control of command paths, not self-sufficiency.[2,3]

BUILD / PRESERVE

Preserve uranium enrichment, on 38% of world exports of enriched uranium, wind parts, and battery components. Catch Up in EVs, EV drivetrain components and battery cells and packs, with allied makers where the base is thin. Judge the nuclear parts and reactor Catch Up calls on reactor-specific measures.[2]

PARTNER

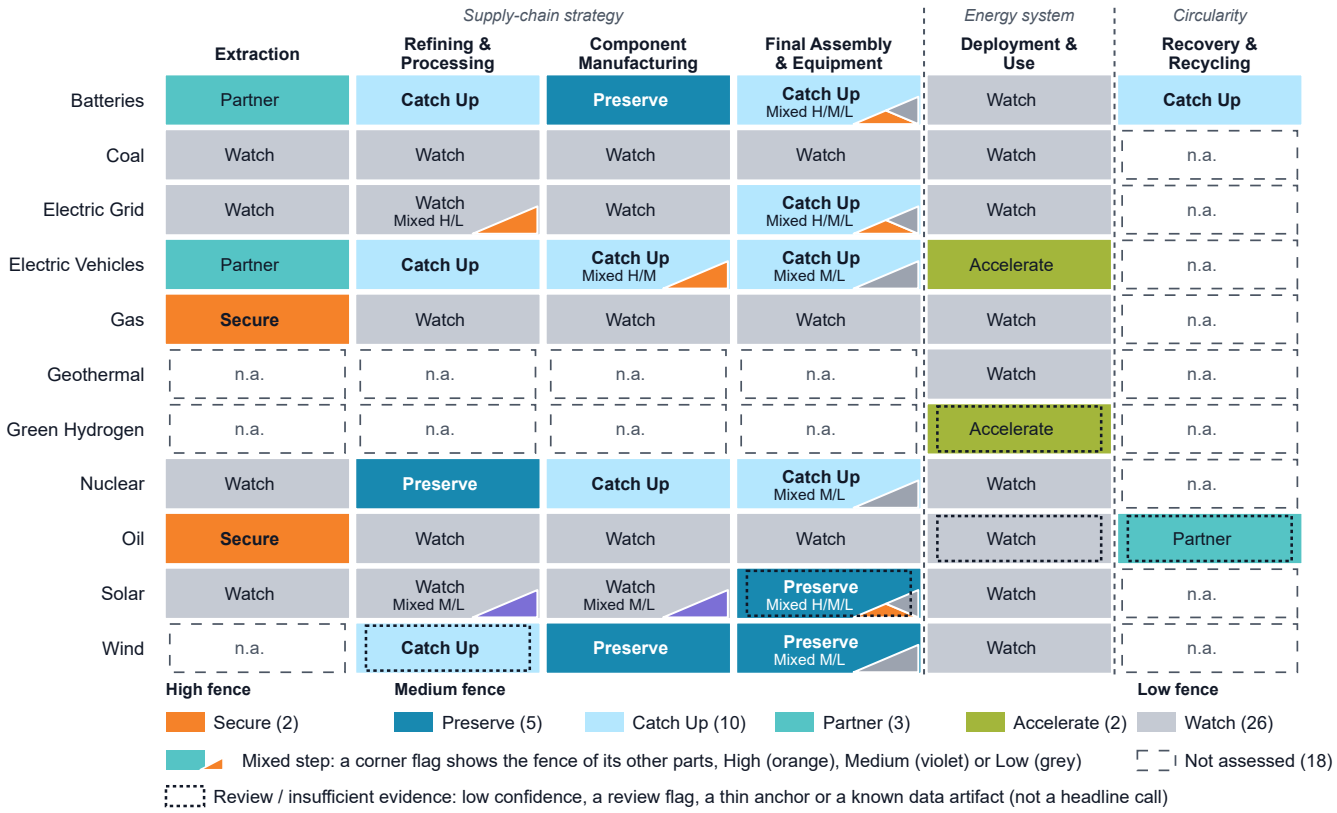
Battery and EV minerals from trusted producers such as Canada, the United States and Australia; allied supply covers the need. With developing partners, add trusted battery-cell capacity through development finance and offtake.[1,2,4]

DEPLOY / STAY OPEN

Accelerate EV adoption and charging, buying equipment competitively from trusted suppliers, with security conditions on connected equipment. Leave open wafers and PV cells, polysilicon, plant equipment for gas, oil and coal, and uranium ore, bought from a diversified base of trusted producers.

Figure 1 shows a manufacturing economy. Preserve and Catch Up fill the component and assembly columns of batteries, EVs, nuclear and wind, and enrichment is the only Preserve among the refining steps. Secure sits only in oil and gas, Partner in the minerals France does not mine, and deployment is mostly Watch. Orange corner flags mark High-fence parts inside other steps: control software, EV magnets and transformer steel. Violet flags mark Catch Up layers inside Watch steps, such as solar cells and polysilicon, which inherit their steps' anchors.

Figure 1. Where France should secure, build, partner or stay open



Each cell is a supply-chain step of a technology, colored by its strategic choice. A corner flag marks a mixed step whose named parts sit behind a different fence: in battery cells and packs, the battery software is Secure, the cells Catch Up and the enclosures Watch. A dotted outline marks a review case, a call resting on low confidence, a review flag, a thin anchor or a known data artifact; this brief does not turn those calls into recommendations. Steps marked n.a. lack the data for a call.[4]

2. Where France should secure, compete and stay open

High fence — what cannot fail

Secure covers two whole steps and eight layers inside other steps, grouped into five capabilities, all in the table with oil and gas together. Secure means surviving a prolonged cutoff or a hostile command through trusted supply, stocks, redundancy or control of the command path, not making everything at home.[4]

CAPABILITY	WHY IT MATTERS	MAIN VULNERABILITY	RECOMMENDED RESPONSE
Oil and gas supply	Need is 98 for oil and 91 for gas out of a hundred, and rebuilding supply takes one to three years.	Potentially coercive suppliers provide 51% of oil and 25% of gas imports; allied supply falls short for oil.	**Diversified and allied.** Stocks sized to the rebuild time; capacity reservations with trusted producers such as Canada, Norway and the United States.
Control software of grids, inverters and batteries	Firmware and grid-control software can command equipment from abroad.	China supplies 51% of battery and 25% of PV and inverter imports: hardware shares, not software.	**Mixed.** Trusted-source and certification rules for firmware and remote access; a replacement program for the installed base.
Grain-oriented electrical steel	The core steel of power transformers.	The call rests on its step, grid metals refining: need 73, trusted suppliers 5% of world supply, a figure unrelated steps repeat; potentially coercive suppliers provide only 6% of its imports.	**Trusted supply.** Capacity reservations with trusted steelmakers; spares pools for transformers.

CAPABILITY	WHY IT MATTERS	MAIN VULNERABILITY	RECOMMENDED RESPONSE
Rare-earth magnets for EV motors	They drive the traction motors of France's EVs.	China holds 94% of rare-earth and battery-metal refining, a step-wide share that battery materials repeats; trusted suppliers 4%; rebuilding takes more than three years.	**Qualified supply.** Offtake and allied agreements; stockpiles; adversary-source limits, phased in as trusted supply qualifies.

Sources: 1, 2, 3, 4.

Turbine, vehicle, reactor and oil and gas network controls read Watch for France. Inside blades, towers and nacelle parts, the same magnets read Preserve on that step's position, not on magnet output.[4]

Medium fence — where industrial capability matters

This is where France makes its real industrial choices: is a capability worth having (Strategic Industrial Value), and how strong is France today (Competitive Position)? Most French positions rest on world supply shares of a few percent, ranking high only among the economies that already make them.

Preserve. Uranium conversion and enrichment reads Preserve on a position of 95. Its measured anchor is exports of enriched uranium, 38% of the world's. Its supply share of 25% counts uranium compounds, yellowcake included, and converted uranium trades under the mining step's codes, so the data see enrichment, not conversion. With Russia holding 22% of that compound-based supply, trusted enrichment matters to allies. Wind turbines (90) and blades, towers and nacelle parts (83) rest on operating capacity, but on 2% of world supply each, and the turbine step's trade codes are mostly cables, transformers and switchgear. Battery components (82) rest on three anchors with moderate confidence. PV modules and inverters read Preserve only on low confidence, a review case.[2,4]

Catch Up. EV and charger manufacturing (62) and EV drivetrain components (54) combine high value with high confidence; vehicles rest on three anchors, operating capacity among them. Battery cells and packs add high need (79) but rest on a single anchor, 1% of world supply, while China provides 51% of imports. Nuclear fuel and reactor parts (71) and reactors and steam generators (37) also read Catch Up, but their trade baskets mix reactor parts with generic boiler and steam-turbine parts. The AIB's own measure, 20% of world exports or production of reactor parts, is the firmer sign of a nuclear supplier.[1,2,4]

Other Catch Up calls are weaker than they look, and this brief does not build on them:

- **Battery materials and rare-earth refining:** both at a position of 0, each on a single export share (2% and 1%). Both show China with 94% of world supply, an identical share that suggests overlapping trade codes.[2,4]
- **Battery recycling:** a single export share (3%) in trade codes for battery waste, scrap and residues.[2]
- **Rare-earth and wind materials refining:** low confidence, a review case.[4]

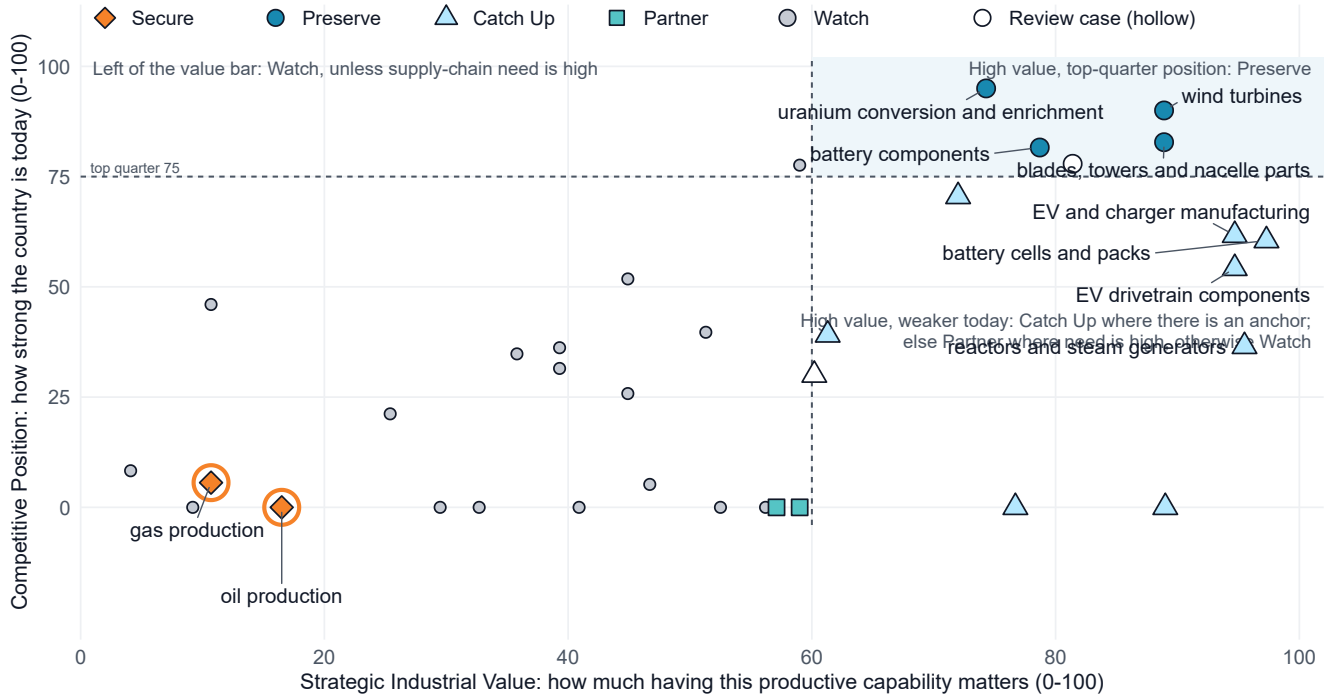
Partner. France holds 0% of world supply in battery and EV minerals and no pathway to scale it, so it should depend deliberately on trusted producers such as Canada, the United States and Australia; allied supply covers the need. Its largest supplier today, Gabon, is outside that list. Waste-oil recovery reads Partner on low confidence, a review case.[1,2,4]

Shared capabilities. Some capabilities sit in several technology chains with different calls; the table keeps the national reading apart from what allies rely on France for.

SHARED CAPABILITY	STEPS AND THEIR CHOICES	NATIONAL READING	AIB ROLE
Uranium conversion & enrichment	uranium mining (Watch) and uranium conversion and enrichment (Preserve)	The choices differ within the same technology with each step's own measures (uranium mining: need 98, value 33, position 0; uranium conversion and enrichment: need 61, value 74, position 95)	Supplier: 10% of world exports or production of uranium conversion & enrichment (the AIB measure of the product, not a step's share)
Accumulator parts & separators	battery components (Preserve) and EV drivetrain components (Catch Up)	The choices differ across technologies with each step's own measures (battery components: need 65, value 79, position 82; EV drivetrain components: need 38, value 95, position 54)	Allies need new capacity here (BUILD); no supplier role
Electricity meters	transformers, switchgear and cables (Catch Up); PV modules and inverters (Preserve, review: low confidence); and wind turbines (Preserve)	The choices differ across technologies with each step's own measures (transformers, switchgear and cables: need 22, value 61, position 39; PV modules and inverters: need 30, value 81, position 78; wind turbines: need 22, value 89, position 90)	Allies need new capacity here (BUILD); no supplier role
Ferrite & other magnets	EV drivetrain components (Catch Up) and blades, towers and nacelle parts (Preserve)	The choices differ across technologies with each step's own measures (EV drivetrain components: need 38, value 95, position 54; blades, towers and nacelle parts: need 34, value 89, position 83)	Allies need new capacity here (BUILD); no supplier role

Sources: 1, 4.

Figure 2. Which producing capabilities are worth holding, and where France is strong today



Producing steps with both measures (recovery has no position measure). Dashed lines mark the value bar (60) and the top quarter (75). Enrichment and the wind steps sit in the upper right; battery materials and rare-earth refining sit on the bottom axis, with no measurable position. Wafers and PV cells hold a top-quarter position (78) but read Watch on low need and a value just under the bar. Hollow markers are review cases.[2,4]

Low fence — what not to prioritize

A credible strategy also says what not to localize. Watch is not a failure: it means affordable, diversified access matters more than domestic production.

- **Wafers and PV cells, and polysilicon refining.** Need for wafers and cells is low (19), their trade codes include chip-industry codes, and their cell layers read Catch Up only on inherited anchors. France has no measurable position in polysilicon refining (0), and its solar-grade layer's Catch Up rests on the step's export share alone.[2,4]
- **Gas, oil and coal plant equipment.** Need is low, value below the bar and positions below the top quarter. Yet 19 economies count France as their best-placed trusted partner for coal-plant equipment, where it holds 9% of world supply: a supplier role to keep reliable through competition, not a fence.[2,4]
- **Uranium mining.** Need is very high (98), and Kazakhstan, the largest potentially coercive supplier, provides 31% of imports. Yet the step reads Watch, a large, commoditized yard where France has no mines: buy from a diversified base of trusted producers.[2,4]

3. France's allied bargain

What France contributes. The Allied Industrial Base analysis assigns France no primary strategic role at its current thresholds. It holds a role in nine of the ten strategic compacts, most of them substitutable. Its supply is essential where it holds core seats, in Nuclear Fuel & Reactor Systems and Electric Vehicles: 20% of world exports or production of reactor parts and 8% of electric buses, trucks and tractors. It also supplies 10% of uranium conversion and enrichment products, natural uranium compounds included. 20 economies count France as their best-placed trusted partner somewhere in the chain, 19 of them for coal-plant equipment.[1,2,4]

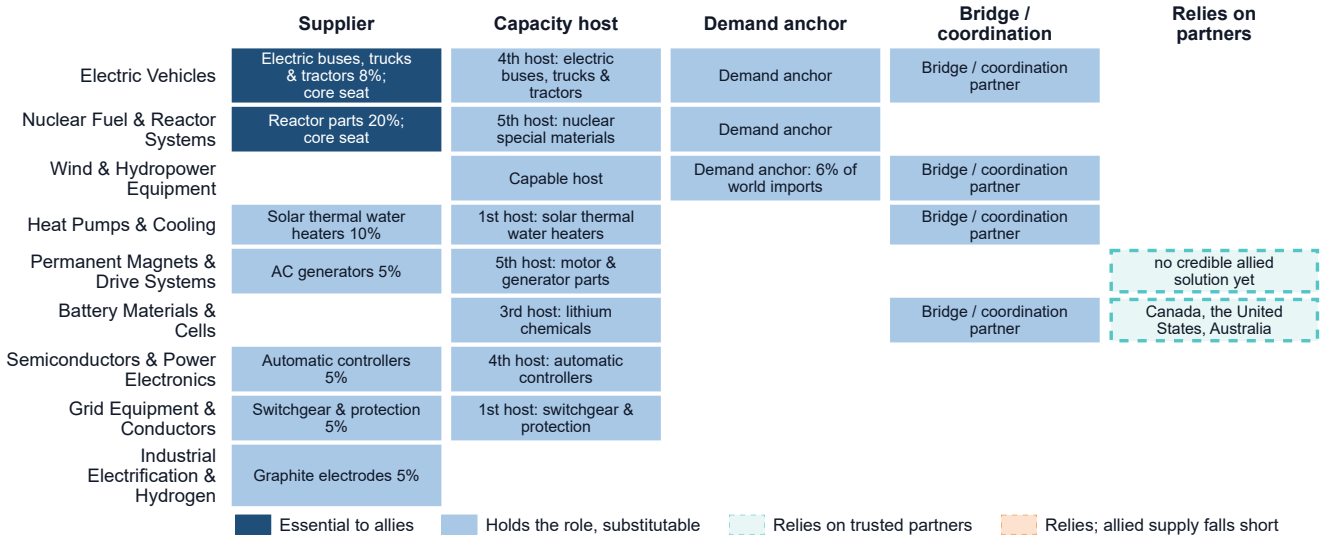
What France needs. Fuel, minerals, magnets and battery cells. If China's supply stopped, as in the Decoupling scenario, France would lose US$8.3 billion a year of imports across the Allied Industrial Base basket. New capacity at allied hosts, led by Germany, would restore only about US$2.6 billion of it a year. Under Managed Interdependence, the default scenario, France sits in the minimum coalition of two Sectoral Compacts (the finer units beneath the strategic compacts); under Decoupling, in five.[1,2]

The dependencies to make intentional. These bargains turn that exchange into a specialization partners can count on.

CAPABILITY / COMPACT	FRANCE CONTRIBUTES	FRANCE NEEDS	NATURAL PARTNERS	WHAT MAKES THE BARGAIN WORK
Nuclear Fuel & Reactor Systems	Reactor parts, the lead product (20% of world exports or production) and enrichment (38% of world exports); a core seat	Uranium and fuel materials; uranium imports led by Kazakhstan	Australia, Canada and India, the compact's resource and fuel seats	Diversified uranium imports; public finance and allied investment for enrichment, tied to benchmarks
Electric Vehicles	Electric buses, trucks and tractors, the lead product (8% of world exports or production); a core seat	Battery cells: China supplies 51% of imports	Germany, the United States, Japan and South Korea; for cells, South Korea	Co-investment, offtake and joint ventures with allied cell makers; time-limited support tied to benchmarks
Battery Materials & Cells	Bridge and diversification roles; hosting (3rd host: lithium chemicals)	Battery and EV minerals; refined materials	Canada, the United States and Australia for minerals; Japan, Indonesia and South Korea for processing	Offtake and co-investment with the recommended partners; development finance for partner capacity
Permanent Magnets & Drive Systems	AC generators, the lead product (5% of world exports or production), not magnets	Qualified magnets: the AIB finds no credible allied solution yet	Germany, the United States, Japan and Mexico, the compact's makers	Offtake and allied agreements; stockpiles until supply qualifies

Sources: 1, 2, 4.

Figure 3. France's role in the Allied Industrial Base



Each row is a strategic compact in which France holds a role or relies on partners. Dark cells are roles allies cannot easily replace; light cells are roles others could fill; dashed cells mark where France relies on trusted partners, orange where allied supply still falls short of what allies need.

Figure 3 shows a supplier with many substitutable roles and essential ones in nuclear and vehicles. France relies on partners in battery minerals, where allied supply covers the need, and in magnets, where no credible allied solution exists yet. In magnets the bargain has to add capacity, not only redirect trade.

4. Development strategy — build trusted capacity by helping partners Catch Up

Development cooperation can be the development arm of that bargain. A partnership pairs a step where France or its allies need more trusted capacity with a developing partner whose same step is a credible Catch Up with a corroborated anchor, inside the allied trust screen. 49 pairs qualify, with 16 developing partners across 10 steps.[1,4,5]

France's own need is a thin base for them. Only 5 pairs, with 5 partners, rest on it, all in battery cells and packs. The other 44 rest on the allied gap alone, a reason that is the same for every donor in scope: they show where allies fall short, not what France needs.[1,4,5]

CAPABILITY / ALLIED GAP	DEVELOPING PARTNER	WHY THIS PARTNER CAN CATCH UP	WHAT THE DONOR SHOULD SUPPORT	SECURITY / AIB BENEFIT	DEVELOPMENT BENEFIT
Battery cells and packs: high need and new allied capacity needed	Malaysia (upper middle income)	A share of world supply and a share of world exports; corroborated by feasibility, input costs, specialization in the adjacent step and industrial learning	Constraint: no operating or committed plant in the capacity data and a position that rests on trade or output shares, not on plant capacity. Lead with development finance, guarantees, offtake, equity and joint ventures; public support for project preparation and investment promotion	Diversified supply for a high-need step and new allied capacity where the AIB finds a gap (allies need new capacity in 2 product groups of the step)	Factory capacity, skills and export sales of finished equipment

CAPABILITY / ALLIED GAP	DEVELOPING PARTNER	WHY THIS PARTNER CAN CATCH UP	WHAT THE DONOR SHOULD SUPPORT	SECURITY / AIB BENEFIT	DEVELOPMENT BENEFIT
Battery materials refining: new allied capacity needed	Madagascar (low income)	A share of world exports and operating capacity; corroborated by specialization in the adjacent step	Constraint: related industrial know-how outside the top quarter and input costs outside the cheapest quarter. Workforce development, technology transfer and joint ventures, industrial infrastructure, feasibility studies and project preparation; then, at scale, offtake, market access, investment guarantees	New allied capacity where the AIB finds a gap (allies need new capacity in 6 product groups of the step)	Processing capacity at home, with the skills, standards and value capture of a refinery
EV and charger manufacturing: new allied capacity needed	Mexico (upper middle income)	A share of world supply, a share of world exports, operating capacity and a strong position in the next step up the chain; corroborated by feasibility, specialization in the adjacent step and industrial learning	Constraint: input costs outside the cheapest quarter. Lead with development finance, guarantees, offtake, equity and joint ventures; public support for industrial infrastructure and supplier development	New allied capacity where the AIB finds a gap (allies need new capacity in 4 product groups of the step)	Factory capacity, skills and export sales of finished equipment
EV drivetrain components: new allied capacity needed	Thailand (upper middle income)	A share of world supply and a share of world exports; corroborated by feasibility, input costs, specialization in the adjacent step and industrial learning	Plant capacity not covered by the data. Lead with development finance, guarantees, offtake, equity and joint ventures; public support for supplier development and technical and vocational skills	New allied capacity where the AIB finds a gap (allies need new capacity in 7 product groups of the step)	A certified supplier base and export sales of components
Wind turbines: new allied capacity needed	Viet Nam (lower middle income)	A share of world supply and a share of world exports; corroborated by feasibility and industrial learning	Constraint: no operating or committed plant in the capacity data and a position that rests on trade or output shares, not on plant capacity. Input costs not covered by the data. Project preparation, investment promotion, industrial infrastructure, quality systems and certification; then, at scale, development finance, guarantees, offtake, equity and joint ventures	New allied capacity where the AIB finds a gap (allies need new capacity in 2 product groups of the step)	Factory capacity, skills and export sales of finished equipment

Sources: 1, 2, 4, 5, 6, 7.

The table leads with the own-need row, Malaysia, an upper middle income economy with a position of 66 but no operating or committed plant in the capacity data. The other rows rest on the allied gap. Finance, guarantees and offtake should lead for Mexico and Thailand. Madagascar (low income, Competitive Position 0) and Viet Nam (lower middle income; its anchor is a trade share on the step's cable-heavy codes, with no plant in the capacity data) need project preparation, infrastructure and skills first.[1,2,4,5]

A partnership lasts longer when it rests on more than aid. The second table shows today's ties: aid is the 2020 to 2024 total at the step, and the direct investment position is economy-wide. Aid at these steps is small or absent, and where the factory-project data cover a pair, they record no French project in the technology.[5]

PARTNERSHIP	RELATIONSHIP TODAY	WHAT MAKES IT DURABLE
Malaysia: battery cells and packs	No energy, industry or mining aid recorded for the pair; a direct investment position of $1,217 million (economy-wide); no factory projects by firms from France recorded in the technology; and no export-credit or lender finance recorded at the step	Long-term offtake, joint ventures and investment guarantees as the capability scales; the partner's role as Specialized technology supplier in the Battery Materials & Cells compact
Madagascar: battery materials refining	$2.0 million of energy, industry and mining aid, none at this step; a direct investment position of $227 million (economy-wide); and no export-credit or lender finance recorded at the step	Long-term offtake, joint ventures and investment guarantees as the capability scales; the partner's role as Resource supplier in the Battery Materials & Cells compact
Mexico: EV and charger manufacturing	$104 million of energy, industry and mining aid, none at this step; a direct investment position of $5,073 million (economy-wide); no factory projects by firms from France recorded in the technology; and an AIB simulated investment pairing (a model result, not an existing tie)	Long-term offtake, joint ventures and investment guarantees as the capability scales; the partner's role as Scale manufacturer in the Electric Vehicles compact; France's own deployment demand for the technology (Accelerate)
Thailand: EV drivetrain components	$0.6 million at this step, under stage-wide codes (the technology is not identified); a direct investment position of $2,453 million (economy-wide); and no factory projects by firms from France recorded in the technology	Long-term offtake, joint ventures and investment guarantees as the capability scales; the partner's role as Resource supplier in the Permanent Magnets & Drive Systems compact; France's own deployment demand for the technology (Accelerate)
Viet Nam: wind turbines	$154 million of energy, industry and mining aid, none at this step; a direct investment position of $1,040 million (economy-wide); and no factory projects by firms from France recorded in the technology	Long-term offtake, joint ventures and investment guarantees as the capability scales; the partner's role as Demand anchor in the Wind & Hydropower Equipment compact; the partner's own deployment demand for the technology (Accelerate)

Sources: 1, 4, 5, 7, 8, 9, 10.

In battery cells and packs, 5 developing economies have credible pathways, led by Malaysia and Indonesia, and Viet Nam already hosts new capacity commercially. For battery mineral mining, EV mineral mining, gas production and oil production: no developing partner passes the tests. Nor does any developing partner qualify in rare-earth and battery-metal refining, where French need is high. Malaysia and Thailand host its new capacity commercially; the other candidates rest on battery-metal capacity data or a thin anchor. Another 29 pairs are commercial, where trade, offtake and investment fit better than aid.[1,4,5]

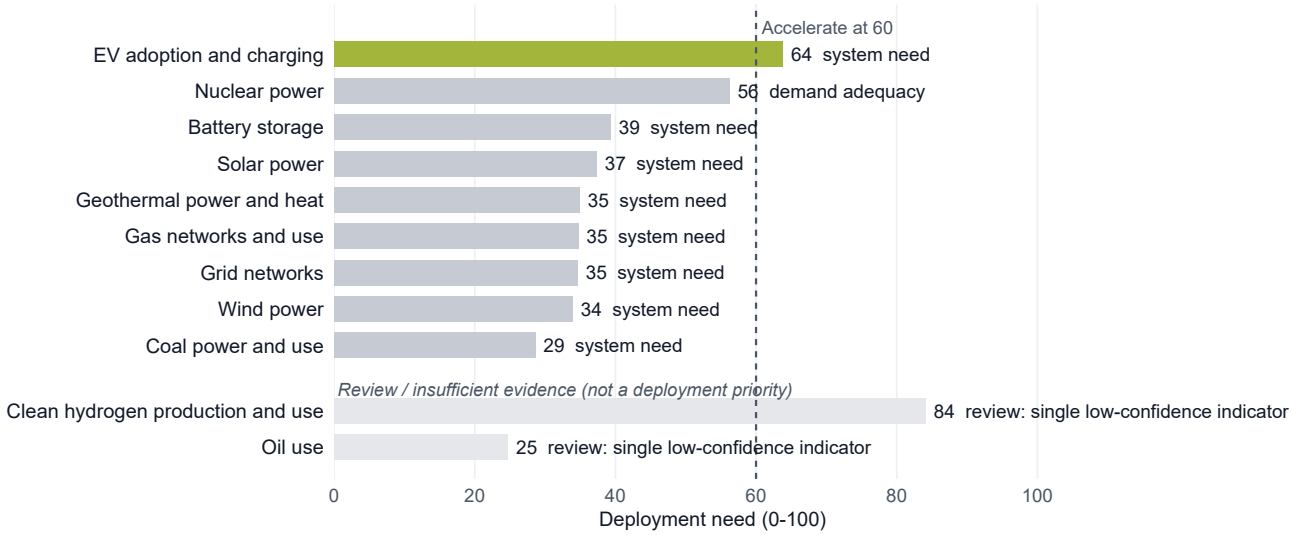
Today's aid does not follow this map; the data cover energy, industry and mining aid only. 7 partners receive French aid at a partnership's step, only under stage-wide codes that do not identify the technology; 8 receive it at other steps, and 2 at none; a partner with several steps can sit in two groups. The largest recipient, South Africa ($434 million over 2020 to 2024), has a credible pathway in battery materials refining, where its aid is coded only by stage. Other large recipients, such as Egypt, Nigeria, Kenya, Cote d'Ivoire, Guinea and Tanzania, have no qualifying partnership here; that aid serves aims this brief does not judge.[1,4,5]

5. Energy-system priorities

Wherever the equipment is made, only EV adoption and charging clears the Accelerate bar of 60 on settled evidence: a need of 64 on two measures with moderate confidence. Clean hydrogen reads higher, but on a single indicator of low confidence, so it is a review case. Nuclear power sits just below the bar (56), though demand adequacy alone (62) clears it and capacity was flat (61.4 to 61.4 GW); it deserves the next review. Renewables are

close to target: 91 GW by 2030 against 84 GW in 2025, a shortfall of about 3 GW at the recent build rate. Storage, solar, wind and grid networks are worth watching, not forcing.[4,11,12]

Figure 4. What France needs to deploy faster



Deployment need by technology, with the largest of its measures: demand adequacy, the gap to national targets, or system need. Review cases, which rest on a single indicator of low confidence, are shown apart in grey and are not deployment priorities.

EV adoption should not wait for the domestic Catch Up. Oil use is a review case on a single indicator of low confidence, so it sets no priority.

6. Is current policy pointed in the right direction?

Recorded policy never decides a choice; it is compared with it afterwards, within France's own measures.

CAPABILITY	STRATEGIC CHOICE	CURRENT POLICY	GAP
Oil production	Secure	Needs more attention	Vulnerability without enough policy attention
Gas production	Secure	Needs more attention	Vulnerability without enough policy attention
Rare-earth and battery-metal refining	Catch Up	Needs more attention	Capability case without enough policy attention
Battery storage	Watch	Heavy local-content protection on the equipment of a Watch deployment	Localizing what is not a domestic priority
Wind power	Watch	Heavy local-content protection on the equipment of a Watch deployment	Localizing what is not a domestic priority

Sources: 4, 13, 14.

First, oil and gas are Secure but rank low in France's own measures (25 and 36): little recorded policy attention; stocks are not measured here, so test stocks and contracts against the rebuild time. Second, rare-earth refining needs attention, but on a small export base, so qualified magnet supply comes first. Finally, local-content protection on wind and storage equipment ranks near the top (99 and 94), though both deployments read Watch. The equipment steps are Preserve and Catch Up: Catch Up allows time-limited local-content rules tied to benchmarks, while Preserve calls only for targeted remedies and conditional procurement.[4,13,14]

7. Strategic agenda — what should France do next?

1. Secure fuel supply and the grid's command layer

Size oil and gas stocks to the rebuild time. Reserve capacity with trusted producers such as Canada, Norway and the United States for oil and Norway, Australia and the United States for gas; partner evidence is weak, so test the terms. Apply trusted-source and certification rules to battery, inverter, metering and grid-control software, and reserve transformer steel with trusted steelmakers.[1,2,4]

2. Keep enrichment competitive as an allied supply, and diversify uranium

Keep enrichment, the nuclear position the data measure most directly, at the frontier with public finance and allied investment tied to benchmarks, so allies can move away from Russia's supply. Buy uranium from diversified trusted producers.[2]

3. Catch up in EVs, drivetrains and battery cells, with support tied to benchmarks

Tie production incentives, workforce development and demand creation to benchmarks, with sunsets. Build at home where the base is established and with allied makers such as Germany, the United States, Japan and South Korea for the rest. Source magnets through offtake and allied agreements that add capacity, since the AIB finds no credible allied solution yet; hold stocks until trusted supply qualifies.[1,4]

4. Build trusted battery-cell capacity with developing partners

Start with Malaysia, the own-need partner, checking first why the capacity data show no plant; lead with development finance, guarantees and offtake. Pursue allied-gap pairs, such as Mexico in EV and charger manufacturing, jointly with allies.[1,4,5]

5. Accelerate EV adoption while keeping sourcing open

Deploy faster through charging infrastructure, permitting and procurement, buying from trusted suppliers with security conditions on connected equipment. Keep local-content protection on storage equipment time-limited and tied to Catch Up benchmarks, with sunsets, and hold wind equipment to targeted remedies and conditional procurement.

Sources

1. Allied Industrial Base model: roles, compacts, trade restored and new capacity across the allied coalition, computed by the Explorer's authors (run of 30 October 2026); CC BY 4.0.
2. United Nations, UN Comtrade Database, HS6 bilateral trade (all reporters); reconciled by the authors (2022-2024 data); attribution; https://comtradeplus.un.org/.
3. Yards & Fences node attributes: standardization, learning, industrial commons and control layers, curated by the authors from public sources (2026-10-30); CC BY 4.0.
4. Country strategy choices by technology and supply-chain step (supply-chain need, Strategic Industrial Value, Competitive Position, deployment need, policy alignment), computed by the Explorer's authors (dashboard data of 31 October 2026); CC BY 4.0.
5. OECD Creditor Reporting System: official development assistance by donor, recipient and purpose code, gross disbursements at constant prices, energy, industry and mining purpose codes only, mapped to supply-chain steps by the authors (Components and Assembly share purpose codes; no code funds recycling) (2020-2024 data); attribution; https://data-explorer.oecd.org/.
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14. International Energy Agency, Policies and Measures Database (export of 2026-10-26); attribution; https://www.iea.org/policies.

Appendix — evidence and method

How this brief is built. Every number comes from the Country page data of the Allied Industrial Base Explorer (the data release of 30 October 2026) and from the Allied Industrial Base analysis. Each technology is split into supply-chain stages from extraction to recycling, and each step gets a strategic choice by fixed rules applied in order:[4]

- Secure where an acute vulnerability is measured;
- at deployment, Accelerate where deployment need reaches 60;[4]
- Watch where the step is a large, commoditized yard;
- Preserve where Competitive Position is in the top quarter (75) and either Strategic Industrial Value (bar 60) or supply-chain need (bar 67) is high;[4]
- Catch Up where the value is high, the position is not, and a capability anchor is corroborated by other evidence;
- Partner where supply-chain need is high but there is neither a top-quarter position nor, for a high-value step, a credible pathway;
- Watch otherwise.

Named layers inside a step, such as control software or magnets, take their own choice, and a step whose parts sit behind different fences is mixed. A value that rests on market size alone is capped just under the bar. Observed policy never decides a choice. The high fence (Secure) protects what cannot be allowed to fail; the medium fence covers capabilities worth building, keeping or sourcing deliberately and deployment to accelerate; the low fence (Watch) leaves a step open to competitive purchase.

Review cases. Some calls rest on low confidence, an engine review flag, a thin capability anchor, a known data artifact or, at deployment, a single indicator of low confidence. They stay on the map but never enter the Bottom line, the Agenda, the policy table or a headline mark. 5 steps are review cases.[4]

STEP	CHOICE	WHY IT IS A REVIEW CASE
Clean hydrogen production and use	Accelerate	A single indicator of low confidence
Oil use	Watch	A single indicator of low confidence
Waste-oil recovery	Partner	Low confidence
PV modules and inverters	Preserve	Low confidence
Rare-earth and wind materials refining	Catch Up	Low confidence

Sources: 4.

Evidence and caveats. The oil and gas calls rest on measured import dependence and supplier concentration, and the EV and battery-cell Catch Up calls have high confidence. The enrichment Preserve rests on exports of enriched uranium; its supply share of 25% is a share in uranium compounds, yellowcake included, and the AIB's measure of 10% also covers natural uranium compounds. Natural uranium hexafluoride, the output of conversion, trades with yellowcake under the mining step's codes, so conversion has no measure of its own. The baskets of nuclear fuel and reactor parts and of reactors mix reactor parts with generic boiler and steam-turbine parts, the kind of equipment that also fills the coal-plant steps. The wind turbines step's trade codes are mostly cables, transformers and switchgear; battery recycling's are battery waste, scrap and residues; rare-earth and wind materials refining's are mostly copper wire rod by value. The solar wafer and module baskets include chip-industry codes, and the Catch Up layers inside the Watch solar steps inherit their steps' anchors. Rare-earth and battery-metal refining and battery materials refining show the identical China share of world supply, which suggests overlapping trade codes. Grain-oriented electrical steel reads Secure on the measures of its step, which cover grid metals as a whole; sintered magnets read Secure in EV drivetrains and Preserve in wind components, each on its step's measures. Uranium mining reads Watch as a large yard despite a need of 98. 18 steps lack the data for a call, including geothermal supply, green hydrogen manufacturing and most recycling. Recorded policy ranks electrolyser systems (96) and stacks (83) near the top of France's own measures, but those steps cannot be assessed and hydrogen deployment need rests on a single indicator, so this brief does not settle that disagreement. Import shares follow the trading partner, so a hub such as Belgium for gas can stand for the producers behind it. Competitive Position ranks France among economies that already supply a step, so a zero means no measurable position. Partner lists name where allied capacity sits, which can include trading hubs, not who owns it.[1,2,4,13,14]

Development partnerships. A pair joins a French step with a strategic reason for more allied capacity and a developing economy (not high income) whose same step is a Catch Up. The reason is France's own need, or the allied gap alone, where allies need new capacity, which is the same for every donor in scope. The partner needs an anchor corroborated beyond

context signals, sufficient evidence and confidence, and a place inside the AIB trust screen; on a step whose data describe different products, the anchor must measure the product the need is about. A partner that already hosts the step's new capacity in the AIB, or Preserves it, is a commercial case. Aid comes from the OECD Creditor Reporting System by purpose code and covers energy, industry and mining aid only. Components and Assembly share codes, no code funds recycling, and many codes name a stage rather than a technology, so aid at a step is not aid for that technology. Aid is called technology-coded only where codes naming the technology hold at least half of it, and a partner with several steps can sit in several aid groups. A constraint on a partner is named only where the data observe its input; otherwise the table says the data do not cover it. In the relationship table an observed zero is stated as nothing recorded, data that do not cover a pair are left out, and the direct investment position is economy-wide, not tied to the step. Export-credit finance is reported apart and never counted as aid, and direct investment uses the IMF positions (the partner's reports where France reports none). The full list of pairs, with every field, is in the development-partnership export.

STEP	DONOR REASON	DEVELOPMENT PARTNERS	COMMERCIAL ROUTE	REVIEW CASES	AID AT THE STEP
Battery cells and packs	High need and new allied capacity needed	Malaysia, Indonesia, Mexico, India and Turkiye	Viet Nam	–	India
Battery materials refining	New allied capacity needed	Madagascar, the Philippines, India, South Africa, Thailand, Papua New Guinea and Turkiye	Argentina, Indonesia, the Democratic Republic of Congo and Malaysia	Mozambique, Colombia and Tanzania	India, South Africa, Thailand, Papua New Guinea and Argentina
EV and charger manufacturing	New allied capacity needed	Mexico, Viet Nam, Thailand, Malaysia, Morocco and Indonesia	Turkiye and India	–	Viet Nam, Thailand and India
EV drivetrain components	New allied capacity needed	Thailand, Morocco, Malaysia, Serbia, Turkiye, Honduras and Ukraine	Indonesia, the Philippines, Mexico and India	Viet Nam	Thailand, Serbia, India and Viet Nam
Blades, towers and nacelle parts	New allied capacity needed	Thailand, Serbia, Turkiye, Brazil and Malaysia	Mexico, Viet Nam, the Philippines and India	–	Thailand, Serbia, Brazil and India
Wind turbines	New allied capacity needed	Viet Nam, Turkiye, Thailand, Indonesia, Morocco and Malaysia	India and Mexico	Brazil	Thailand, India and Brazil
Battery components	New allied capacity needed	Viet Nam, Indonesia, Turkiye, Mexico, India, Morocco and Malaysia	–	–	India
Transformers, switchgear and cables	New allied capacity needed	India, Malaysia and the Philippines	Mexico, Thailand, Turkiye and Viet Nam	Morocco	India and Thailand
Rare-earth and wind materials refining	New allied capacity needed	India	–	Viet Nam and South Africa	India and South Africa
PV modules and inverters	New allied capacity needed	Morocco and Turkiye	India, Viet Nam, Thailand, Mexico, Malaysia and Indonesia	–	India and Thailand
Rare-earth and battery-metal refining	High need and new allied capacity needed	–	Malaysia and Thailand	Argentina, Indonesia, the Democratic Republic of Congo, Mozambique, the Philippines, Mexico, Madagascar and Tanzania	Thailand and Argentina
Nuclear fuel and reactor parts	High need	–	–	India, Thailand and Viet Nam	India and Thailand
Battery recycling	High need	–	–	India and Malaysia	–

Sources: 1, 4, 5.

Every step.

STEP	STAGE	CHOICE	FENCE	NEED	VALUE	POSITION	EVIDENCE	POLICY
Battery mineral mining	Extraction	Partner	Medium	90	57	0	strong	broadly aligned
Battery materials refining	Refining	Catch Up	Medium	52	77	0	strong	broadly aligned
Battery components	Components	Preserve	Medium	65	79	82	moderate	broadly aligned
Battery cells and packs	Assembly	Catch Up	Mixed High/Medium/Low	79	97	61	moderate	broadly aligned
Battery storage	Deployment	Watch	Low	39	–	–	moderate	too much intervention
Battery recycling	Recovery	Catch Up	Medium	87	82	–	–	mixed signals
Coal mining	Extraction	Watch	Low	91	4	8	strong	broadly aligned
Coke and coal processing	Refining	Watch	Low	87	9	0	strong	broadly aligned
Coal-plant components	Components	Watch	Low	56	39	36	strong	broadly aligned
Coal-plant equipment	Assembly	Watch	Low	38	39	32	strong	mixed signals

STEP	STAGE	CHOICE	FENCE	NEED	VALUE	POSITION	EVIDENCE	POLICY
Coal power and use	Deployment	Watch	Low	29	–	–	moderate	broadly aligned
Coal recovery	Recovery	Not assessed	–	–	1	–	–	not enough policy evidence
Copper and bauxite mining	Extraction	Watch	Low	89	41	0	strong	broadly aligned
Grid metals refining	Refining	Watch	Mixed High/Low	73	53	0	strong	broadly aligned
Grid components	Components	Watch	Low	30	47	5	strong	broadly aligned
Transformers, switchgear and cables	Assembly	Catch Up	Mixed High/Medium/Low	22	61	39	strong	mixed signals
Grid networks	Deployment	Watch	Low	35	–	–	–	broadly aligned
Grid-equipment recycling	Recovery	Not assessed	–	–	14	–	–	not enough policy evidence
EV mineral mining	Extraction	Partner	Medium	90	59	0	strong	broadly aligned
Rare-earth and battery-metal refining	Refining	Catch Up	Medium	87	89	0	strong	needs more attention
EV drivetrain components	Components	Catch Up	Mixed High/Medium	38	95	54	moderate	mixed signals
EV and charger manufacturing	Assembly	Catch Up	Mixed Medium/Low	21	95	62	moderate	broadly aligned
EV adoption and charging	Deployment	Accelerate	Medium	64	–	–	strong	broadly aligned
EV recycling	Recovery	Not assessed	–	–	42	–	–	not enough policy evidence
Gas production	Extraction	Secure	High	91	11	6	strong	needs more attention
Gas liquids processing	Refining	Watch	Low	58	11	46	strong	broadly aligned
Gas-turbine and drilling components	Components	Watch	Low	41	45	52	strong	broadly aligned
Gas turbines and LNG plant	Assembly	Watch	Low	21	45	26	strong	broadly aligned
Gas networks and use	Deployment	Watch	Low	35	–	–	strong	broadly aligned
Gas recovery	Recovery	Not assessed	–	–	7	–	–	not enough policy evidence
Geothermal resource development	Extraction	Not assessed	–	–	20	–	–	not enough policy evidence
Geothermal refining	Refining	Not assessed	–	–	20	–	–	not enough policy evidence
Geothermal plant components	Components	Not assessed	–	–	69	–	–	not enough policy evidence
Geothermal power-plant equipment	Assembly	Not assessed	–	–	69	–	–	not enough policy evidence
Geothermal power and heat	Deployment	Watch	Low	35	–	–	strong	broadly aligned
Geothermal recovery	Recovery	Not assessed	–	–	20	–	–	not enough policy evidence
Electrolyser mineral mining	Extraction	Not assessed	–	–	35	0	strong	not enough policy evidence
PGM and catalyst refining	Refining	Not assessed	–	–	63	91	strong	not enough policy evidence
Electrolyser stacks	Components	Not assessed	–	–	99	–	–	not enough policy evidence
Electrolyser systems	Assembly	Not assessed	–	–	90	–	–	not enough policy evidence
Clean hydrogen production and use	Deployment	Accelerate	Medium	84	–	–	–	broadly aligned
Stack and catalyst recycling	Recovery	Not assessed	–	–	49	–	–	not enough policy evidence
Uranium mining	Extraction	Watch	Low	98	33	0	strong	broadly aligned
Uranium conversion and enrichment	Refining	Preserve	Medium	61	74	95	strong	mixed signals
Nuclear fuel and reactor parts	Components	Catch Up	Medium	80	72	71	strong	broadly aligned
Reactors and steam generators	Assembly	Catch Up	Mixed Medium/Low	43	96	37	strong	broadly aligned
Nuclear power	Deployment	Watch	Low	56	–	–	strong	broadly aligned
Spent-fuel reprocessing	Recovery	Not assessed	–	–	74	–	–	not enough policy evidence
Oil production	Extraction	Secure	High	98	17	0	strong	needs more attention
Oil refining	Refining	Watch	Low	28	36	35	strong	broadly aligned
Drilling and pump components	Components	Watch	Low	41	25	21	strong	broadly aligned
Drilling rigs, tankers and oil equipment	Assembly	Watch	Low	12	51	40	strong	broadly aligned
Oil use	Deployment	Watch	Low	25	–	–	strong	broadly aligned
Waste-oil recovery	Recovery	Partner	Medium	91	8	–	–	broadly aligned
PV mineral mining	Extraction	Watch	Low	93	30	0	strong	broadly aligned
Polysilicon refining	Refining	Watch	Mixed Medium/Low	53	56	0	strong	broadly aligned
Wafers and PV cells	Components	Watch	Mixed Medium/Low	19	59	78	moderate	broadly aligned

STEP	STAGE	CHOICE	FENCE	NEED	VALUE	POSITION	EVIDENCE	POLICY
PV modules and inverters	Assembly	Preserve	Mixed High/Medium/Low	30	81	78	moderate	broadly aligned
Solar power	Deployment	Watch	Low	37	–	–	strong	broadly aligned
PV module recycling	Recovery	Not assessed	–	–	45	–	–	not enough policy evidence
Wind mineral mining	Extraction	Not assessed	–	–	46	8	strong	not enough policy evidence
Rare-earth and wind materials refining	Refining	Catch Up	Medium	64	60	30	strong	needs more attention
Blades, towers and nacelle parts	Components	Preserve	Medium	34	89	83	moderate	broadly aligned
Wind turbines	Assembly	Preserve	Mixed Medium/Low	22	89	90	moderate	broadly aligned
Wind power	Deployment	Watch	Low	34	–	–	strong	too much intervention
Wind-turbine recycling	Recovery	Not assessed	–	–	34	–	–	not enough policy evidence

Sources: 2, 3, 4, 6, 11, 12, 13, 14.

Named layers.

LAYER	STEP	CHOICE	FENCE
Anode materials (natural and synthetic graphite)	battery materials refining	Catch Up	Medium
Cathode active materials and precursors	battery materials refining	Catch Up	Medium
Battery-grade lithium hydroxide and carbonate	battery materials refining	Catch Up	Medium
Battery cells (LFP and NMC)	battery cells and packs	Catch Up	Medium
Communications gateway and cloud fleet management	battery cells and packs	Secure	High
Battery-management firmware, energy-management software and remote updates	battery cells and packs	Secure	High
Enclosures, racking, thermal management and balance of system	battery cells and packs	Watch	Low
Grain-oriented electrical steel	grid metals refining	Secure	High
Power conversion hardware (inverters and converters)	transformers, switchgear and cables	Catch Up	Medium
Smart meters and advanced metering head-end systems	transformers, switchgear and cables	Secure	High
Cables and conductors	transformers, switchgear and cables	Watch	Low
Utility-scale inverter firmware and remote-access functions	transformers, switchgear and cables	Secure	High
Grid-control software (SCADA, EMS, DERMS)	transformers, switchgear and cables	Secure	High
Large power transformers	transformers, switchgear and cables	Catch Up	Medium
HVDC converter-station control and protection	transformers, switchgear and cables	Catch Up	Medium
Electric drive units and motors	EV drivetrain components	Catch Up	Medium
Sintered NdFeB permanent magnets (traction motors)	EV drivetrain components	Secure	High
Connected-vehicle telematics, over-the-air software and connectivity systems	EV and charger manufacturing	Watch	Low
Charging-network management software (charge-point back ends)	EV adoption and charging	Accelerate	Medium
Pipeline SCADA, compressor-station and LNG-terminal control systems	gas turbines and LNG plant	Watch	Low
Iridium and platinum for PEM electrolyser catalysts	electrolyser mineral mining	Not assessed	–
HALEU enrichment	uranium conversion and enrichment	Preserve	Medium
LEU enrichment services	uranium conversion and enrichment	Preserve	Medium
Fuel assemblies (design-specific fabrication)	nuclear fuel and reactor parts	Catch Up	Medium
Reactor control and safety instrumentation	reactors and steam generators	Watch	Low
Oil pipeline and terminal control systems	drilling rigs, tankers and oil equipment	Watch	Low
Solar-grade polysilicon	polysilicon refining	Catch Up	Medium
Solar cells and wafers	wafers and PV cells	Catch Up	Medium
Next-generation PV (perovskite and tandem cells)	wafers and PV cells	Catch Up	Medium
Distributed PV inverter firmware, communication dongles and vendor cloud	PV modules and inverters	Secure	High
Conventional crystalline-silicon modules	PV modules and inverters	Preserve	Medium
Passive racking and mounting	PV modules and inverters	Watch	Low
Rare-earth separation and metal-making	rare-earth and wind materials refining	Catch Up	Medium
Sintered NdFeB permanent magnets	blades, towers and nacelle parts	Preserve	Medium
Turbine controllers, OEM remote access and park SCADA	wind turbines	Watch	Low

Sources: 3, 4.

Partner recommendations.

STEP	CHOICE	TRUSTED PARTNERS	ALLIED SUPPLY	EVIDENCE
Battery mineral mining	Partner	Canada, the United States and Australia	Allied supply covers the need	moderate
EV mineral mining	Partner	the United States, Canada and Australia	Allied supply covers the need	moderate
Gas production	Secure	Norway, Australia and the United States	Allied supply covers the need	weak

STEP	CHOICE	TRUSTED PARTNERS	ALLIED SUPPLY	EVIDENCE
Oil production	Secure	Canada, Norway and the United States	Allied supply falls short	weak
Waste-oil recovery	Partner	No credible allied solution yet	–	none

Sources: 1, 2, 4.

Compact roles.

STRATEGIC COMPACT	LEAD ROLE	FUNCTIONS	LEAD STEP	CORE COUNTRIES
Permanent Magnets & Drive Systems	Scale manufacturer	Scale manufacturer and capacity-building host	Motors & generators	Thailand, Viet Nam, Germany, the United States, Japan and Mexico
Battery Materials & Cells	Bridge / coordination partner	Bridge / coordination partner and diversification partner	–	the United States, the Democratic Republic of Congo, Japan, Indonesia, South Korea and Germany
Semiconductors & Power Electronics	Scale manufacturer	Scale manufacturer and capacity-building host	Converters & controllers	Brazil, Taiwan, Singapore, South Korea, the United States and Viet Nam
Wind & Hydropower Equipment	Demand anchor	Demand anchor, capacity-building host, bridge / coordination partner and diversification partner	Blade materials	Spain, Poland, India, Germany, the United States and the United Kingdom
Nuclear Fuel & Reactor Systems	Specialized technology supplier	Processing anchor, specialized technology supplier, scale manufacturer and demand anchor	Reactor components	Australia, Canada, India, the United States, France and the United Kingdom
Grid Equipment & Conductors	Scale manufacturer	Scale manufacturer and capacity-building host	Network equipment	Chile, Peru, the Democratic Republic of Congo, Japan, Germany and the United States
Electric Vehicles	Scale manufacturer	Scale manufacturer, demand anchor, capacity-building host, bridge / coordination partner and diversification partner	Electric vehicles	Germany, the United States, France, Japan, South Korea and the United Kingdom
Heat Pumps & Cooling	Scale manufacturer	Scale manufacturer, capacity-building host, bridge / coordination partner and diversification partner	Heating & cooling units	the United States, the Netherlands, Thailand, Mexico, Germany and Italy
Industrial Electrification & Hydrogen	Scale manufacturer	Scale manufacturer	Green iron & furnace inputs	India, Oman, South Africa, the United States, Saudi Arabia and Morocco

Sources: 1.

Coverage and freshness.

ITEM	VALUE
Data release	30 October 2026
Steps with a choice	48
Steps not assessed	18
Named layers with a choice	34
Steps with strong position evidence	37
Steps with moderate or weak position evidence	10
Steps with no position evidence	19

Sources: 4.
